

BUSINESS REQUIREMENT DOCUMENT (BRD)

SmartCRM79

Versi Formal untuk Approval Manajemen

Field	Value
Document ID	BRD-SCRM79-001
Project Name	SmartCRM79
Document Type	Business Requirement Document (Formal / Management Approval)
Status	Draft for Management Approval
Prepared By	[Isi nama penyusun / PM / BA]
Reviewed By	[Isi nama reviewer]
Approved By	[Isi nama approver]
Confidentiality	Internal Use Only

Version History

Version	Date	Change Description	Author/Owner
0.1	[Isi tanggal]	Initial formal BRD draft for SmartCRM79	[Isi nama]
0.2	[Isi tanggal]	Management review update	[Isi nama]
1.0	[Isi tanggal]	Approved baseline	[Isi approver]

Approval / Sign-off

Role	Name	Title	Decision (Approve/Revis e)	Signature	Date
Sponsor / Management	[Isi nama]	[Isi jabatan]			
Project Owner	[Isi nama]	[Isi jabatan]			
Business Representative	[Isi nama]	[Isi jabatan]			
Technical Reviewer	[Isi nama]	[Isi jabatan]			

1. Executive Summary

SmartCRM79 adalah proyek pengembangan sistem Customer Relationship Management (CRM) terpadu yang memadukan Operational CRM, Analytical CRM, dan Collaborative CRM dalam satu platform. Solusi ini dirancang untuk mengelola seluruh siklus hubungan pelanggan mulai dari autentikasi pengguna, data pelanggan, sales automation, marketing automation, kolaborasi tim, notifikasi, integrasi API, layanan pelanggan, hingga segmentasi dan prediksi perilaku pelanggan.

2. Latar Belakang

Organisasi modern memerlukan satu platform terpadu untuk mengelola hubungan pelanggan secara konsisten dan real-time.

Dalam praktik, data pelanggan sering tersebar di berbagai file dan tool sehingga riwayat interaksi, status penjualan, campaign marketing, dan ticket layanan tidak terhubung dengan baik.

SmartCRM79 dikembangkan untuk menjadi single source of truth bagi data pelanggan dan aktivitas bisnis lintas fungsi sales, marketing, support, dan manajemen.

3. Maksud dan Tujuan

Maksud: Membangun sistem CRM terintegrasi yang menyatukan proses penjualan, pemasaran, pelayanan pelanggan, kolaborasi tim, dan analitik dalam satu platform.

- Menyediakan single source of truth untuk data pelanggan.
- Meningkatkan efisiensi kerja tim sales, marketing, support, dan manajemen.
- Menyediakan otomatisasi proses bisnis inti CRM.
- Menyediakan dashboard dan laporan analitik untuk pengambilan keputusan.
- Meningkatkan kualitas layanan pelanggan melalui ticketing, SLA, dan feedback.
- Menyediakan kesiapan integrasi dengan channel atau sistem pihak ketiga.

4. Scope

In Scope: Authentication & User Management, Customer Data Management, Sales Automation, Analytics Dashboard & Reporting, Team Collaboration & Communication, Main Dashboard & Notifications, System Integration & API, Marketing Automation, Customer Service System, Customer Segmentation & Prediction

Out of Scope: Integrasi production final ke pihak ketiga, model AI/ML kompleks di luar prediksi sederhana, dan area bisnis di luar domain CRM.

5. User Roles

Role	Primary Responsibility
Super Admin	Mengelola konfigurasi sistem, user, role, dan monitoring keseluruhan.
Sales	Mengelola lead, opportunity, pipeline, dan aktivitas penjualan.
Marketing	Mengelola campaign, template, automation workflow, dan engagement tracking.
Support	Mengelola ticket, SLA, knowledge base, dan feedback pelanggan.
Manager/Analyst	Mengakses dashboard, report, insight, dan monitoring performa bisnis.

6. Global Flow Diagram

Global CRM Flow

[Login & Role Check] -> [Access Dashboard] -> [Manage Customer Data] -> [Sales / Marketing / Support Process] -> [Collaboration & Notification] -> [Analytics & Reporting] -> [Insight / Action / Export]

7. Solution Scope by Module

Req ID	Module	Business Capability
BR-001	Authentication & User Management	Modul keamanan dan kontrol akses yang menjadi fondasi seluruh sistem CRM.
BR-002	Customer Data Management	Core database pelanggan sebagai single source of truth untuk ketiga jenis CRM.
BR-003	Sales Automation	Operational CRM untuk membantu tim sales mengelola proses penjualan dari lead hingga closing.
BR-004	Analytics Dashboard & Reporting	Analytical CRM untuk visualisasi data dan pengambilan keputusan manajemen.
BR-005	Team Collaboration & Communication	Collaborative CRM untuk memastikan seluruh tim bekerja dengan data yang sama dan real-time.
BR-006	Main Dashboard & Notifications	Halaman beranda yang mengintegrasikan seluruh sistem dan memberi overview cepat sesuai role.
BR-007	System Integration & API	Lapisan integrasi backend yang menghubungkan semua modul agar data selalu sinkron.
BR-008	Marketing Automation	Operational CRM untuk membantu tim marketing merencanakan, menjalankan, dan mengukur kampanye secara efisien.
BR-009	Customer Service System	Operational CRM untuk mengelola tiket dan layanan purna jual secara cepat dan terorganisir.
BR-010	Customer Segmentation & Prediction	Analytical CRM untuk mengolah data pelanggan dan memprediksi perilaku dasar.

8. Detailed Business Requirements

BR-001 – Authentication & User Management

Ringkasan: Modul keamanan dan kontrol akses yang menjadi fondasi seluruh sistem CRM.

Requirement Statement: Sistem harus menyediakan kapabilitas Authentication & User Management sesuai fitur baseline yang disepakati untuk SmartCRM79.

Fitur Utama:

- Login/register dengan email/password dan opsi Google OAuth
- Multi-Factor Authentication (MFA)
- Forgot password & reset link via email
- Role default: Super Admin, Sales, Marketing, Support, Manager/Analyst
- Permission granular per modul (view, create, edit, delete, export, analytics)
- Profile management, avatar upload, ubah password
- Audit log aktivitas pengguna
- Session management dan logout all devices
- Dark/light mode toggle

Flow Diagram – Authentication & User Management

```
[Start] -> [Login/Register] -> [MFA] -> [Session/Token] -> [Role & Permission Check] -> [Access Module] -> [Audit Log]
```

Acceptance Criteria (minimum):

- Fitur utama modul dapat didemonstrasikan minimal pada happy path.
- State UI, validasi, dan kontrol akses minimum terdokumentasi.
- Ada bukti integrasi data/aksi sesuai konteks modul.
- Ada dokumentasi handover dan catatan known issues/next steps.

BR-002 – Customer Data Management

Ringkasan: Core database pelanggan sebagai single source of truth untuk ketiga jenis CRM.

Requirement Statement: Sistem harus menyediakan kapabilitas Customer Data Management sesuai fitur baseline yang disepakati untuk SmartCRM79.

Fitur Utama:

- CRUD lengkap dengan validasi data real-time
- 20+ field standar dan unlimited custom fields
- Riwayat interaksi otomatis / activity timeline
- Import/Export CSV, Excel, JSON dengan template
- Advanced search, multi-filter, full-text search
- Duplicate detection & smart merge
- Tagging, segmentation manual, favorite list
- Attachment file dengan preview
- Activity log per pelanggan

Flow Diagram – Customer Data Management

```
[Open Customer Module] -> [Create/Search Customer] -> [View/Edit Profile] -> [Tag/Segment/Attach File] -> [Save] -> [Timeline Updated]
```

Acceptance Criteria (minimum):

- Fitur utama modul dapat didemokan minimal pada happy path.
- State UI, validasi, dan kontrol akses minimum terdokumentasi.
- Ada bukti integrasi data/aksi sesuai konteks modul.
- Ada dokumentasi handover dan catatan known issues/next steps.

BR-003 – Sales Automation

Ringkasan: Operational CRM untuk membantu tim sales mengelola proses penjualan dari lead hingga closing.

Requirement Statement: Sistem harus menyediakan kapabilitas Sales Automation sesuai fitur baseline yang disepakati untuk SmartCRM79.

Fitur Utama:

- Lead & Opportunity management dengan custom stages
- Sales Pipeline Kanban + list view + calendar view
- Task otomatis & reminder
- Deal value, probability, expected close date, forecasting dashboard
- Email template & simulasi pengiriman proposal
- Lead scoring otomatis berdasarkan kriteria
- Win/Loss reason analysis
- Integrasi dengan modul pelanggan & kolaborasi

Flow Diagram – Sales Automation

```
[Create/Import Lead] -> [Qualify Lead] -> [Move Opportunity Stage] ->
[Auto Task & Reminder] -> [Proposal/Negotiation] -> [Win/Loss] ->
[Forecast Updated]
```

Acceptance Criteria (minimum):

- Fitur utama modul dapat didemokan minimal pada happy path.
- State UI, validasi, dan kontrol akses minimum terdokumentasi.
- Ada bukti integrasi data/aksi sesuai konteks modul.
- Ada dokumentasi handover dan catatan known issues/next steps.

BR-004 – Analytics Dashboard & Reporting

Ringkasan: Analytical CRM untuk visualisasi data dan pengambilan keputusan manajemen.

Requirement Statement: Sistem harus menyediakan kapabilitas Analytics Dashboard & Reporting sesuai fitur baseline yang disepakati untuk SmartCRM79.

Fitur Utama:

- Interactive dashboard dengan chart bar/line/pie/funnel
- KPI utama: total revenue, conversion rate, churn rate, customer lifetime value
- Filter berdasarkan tanggal, sales person, produk, region

- Custom report builder
- Export PDF, Excel, CSV dengan branding
- Scheduled report via email
- Role-based dashboard

Flow Diagram – Analytics Dashboard & Reporting

```
[Collect CRM Data] -> [Aggregate KPI] -> [Apply Filters] -> [Render Dashboard/Charts] -> [Export/Schedule] -> [Insight]
```

Acceptance Criteria (minimum):

- Fitur utama modul dapat didemokan minimal pada happy path.
- State UI, validasi, dan kontrol akses minimum terdokumentasi.
- Ada bukti integrasi data/aksi sesuai konteks modul.
- Ada dokumentasi handover dan catatan known issues/next steps.

BR-005 – Team Collaboration & Communication

Ringkasan: Collaborative CRM untuk memastikan seluruh tim bekerja dengan data yang sama dan real-time.

Requirement Statement: Sistem harus menyediakan kapabilitas Team Collaboration & Communication sesuai fitur baseline yang disepakati untuk SmartCRM79.

Fitur Utama:

- Shared customer profile
- Activity feed real-time per pelanggan
- Internal notes, mention, comment thread
- Real-time status update
- Team notification & in-app alert
- Shared calendar untuk meeting/follow-up
- Audit trail lengkap

Flow Diagram – Team Collaboration & Communication

```
[Open Shared Customer Profile] -> [Add Note/Comment/Mention] -> [Status Update] -> [Notification Broadcast] -> [Calendar Follow-up] -> [Audit Trail]
```

Acceptance Criteria (minimum):

- Fitur utama modul dapat didemokan minimal pada happy path.
- State UI, validasi, dan kontrol akses minimum terdokumentasi.
- Ada bukti integrasi data/aksi sesuai konteks modul.
- Ada dokumentasi handover dan catatan known issues/next steps.

BR-006 – Main Dashboard & Notifications

Ringkasan: Halaman beranda yang mengintegrasikan seluruh sistem dan memberi overview cepat sesuai role.

Requirement Statement: Sistem harus menyediakan kapabilitas Main Dashboard & Notifications sesuai fitur baseline yang disepakati untuk SmartCRM79.

Fitur Utama:

- Personalized dashboard berdasarkan role
- Quick links ke semua modul
- Real-time notification center
- Today's task & upcoming deadline
- Mini charts summary
- Welcome message & quick start guide

Flow Diagram – Main Dashboard & Notifications

```
[Login Success] -> [Load Role-based Dashboard] -> [Show Notifications]
-> [Show Tasks/Deadlines] -> [Navigate via Quick Links]
```

Acceptance Criteria (minimum):

- Fitur utama modul dapat didemokan minimal pada happy path.
- State UI, validasi, dan kontrol akses minimum terdokumentasi.
- Ada bukti integrasi data/aksi sesuai konteks modul.
- Ada dokumentasi handover dan catatan known issues/next steps.

BR-007 – System Integration & API

Ringkasan: Lapisan integrasi backend yang menghubungkan semua modul agar data selalu sinkron.

Requirement Statement: Sistem harus menyediakan kapabilitas System Integration & API sesuai fitur baseline yang disepakati untuk SmartCRM79.

Fitur Utama:

- RESTful API lengkap
- Dokumentasi Swagger
- Data synchronization antar modul
- Webhook simulation untuk third-party
- Error logging & monitoring
- Rate limiting & security (JWT)
- Persiapan integrasi eksternal (mis. Gmail, WhatsApp Business API)

Flow Diagram – System Integration & API

```
[Module Event Triggered] -> [API Service] -> [Validation/Security
Check] -> [Sync Module] -> [Log/Monitor] -> [Optional Webhook]
```

Acceptance Criteria (minimum):

- Fitur utama modul dapat didemokan minimal pada happy path.
- State UI, validasi, dan kontrol akses minimum terdokumentasi.
- Ada bukti integrasi data/aksi sesuai konteks modul.
- Ada dokumentasi handover dan catatan known issues/next steps.

BR-008 – Marketing Automation

Ringkasan: Operational CRM untuk membantu tim marketing merencanakan, menjalankan, dan mengukur kampanye secara efisien.

Requirement Statement: Sistem harus menyediakan kapabilitas Marketing Automation sesuai fitur baseline yang disepakati untuk SmartCRM79.

Fitur Utama:

- Campaign builder (email, SMS simulation)
- Email template designer
- Scheduling & recurring campaign
- Automation workflow berbasis trigger
- Tracking open rate, click rate, conversion rate
- Lead scoring dari marketing activities
- A/B testing untuk subject line & content
- Landing page form builder (simulasi)

Flow Diagram – Marketing Automation

```
[Create Campaign] -> [Design Template] -> [Set Segment/Trigger] -> [Schedule/Run] -> [Track Engagement] -> [Update Lead Score]
```

Acceptance Criteria (minimum):

- Fitur utama modul dapat didemokan minimal pada happy path.
- State UI, validasi, dan kontrol akses minimum terdokumentasi.
- Ada bukti integrasi data/aksi sesuai konteks modul.
- Ada dokumentasi handover dan catatan known issues/next steps.

BR-009 – Customer Service System

Ringkasan: Operational CRM untuk mengelola ticket dan layanan purna jual secara cepat dan terorganisir.

Requirement Statement: Sistem harus menyediakan kapabilitas Customer Service System sesuai fitur baseline yang disepakati untuk SmartCRM79.

Fitur Utama:

- Ticketing system dengan nomor tiket otomatis
- Assignment otomatis berdasarkan kategori & beban kerja

- Escalation rules & SLA monitoring
- Knowledge base / FAQ searchable
- Customer feedback & rating
- Simulasi live chat + history chat
- Integrasi dengan data pelanggan & sales pipeline

Flow Diagram – Customer Service System

```
[Create Ticket] -> [Auto Assign] -> [Work on Ticket] -> [SLA Monitor/Escalate] -> [Resolve] -> [Collect Feedback]
```

Acceptance Criteria (minimum):

- Fitur utama modul dapat didemokan minimal pada happy path.
- State UI, validasi, dan kontrol akses minimum terdokumentasi.
- Ada bukti integrasi data/aksi sesuai konteks modul.
- Ada dokumentasi handover dan catatan known issues/next steps.

BR-010 – Customer Segmentation & Prediction

Ringkasan: Analytical CRM untuk mengolah data pelanggan dan memprediksi perilaku dasar.

Requirement Statement: Sistem harus menyediakan kapabilitas Customer Segmentation & Prediction sesuai fitur baseline yang disepakati untuk SmartCRM79.

Fitur Utama:

- RFM Analysis otomatis
- Segmentasi otomatis (High Value, Loyal, At Risk, Lost, dll)
- Customer behavior analysis
- Prediksi pembelian sederhana
- Rekomendasi produk/upsell berdasarkan segment
- Churn prediction dashboard
- Insight card untuk manajemen

Flow Diagram – Customer Segmentation & Prediction

```
[Collect Activity/Transaction Data] -> [RFM Calculation] -> [Segment Assignment] -> [Predict Purchase/Churn] -> [Generate Insight] -> [Recommendation]
```

Acceptance Criteria (minimum):

- Fitur utama modul dapat didemokan minimal pada happy path.
- State UI, validasi, dan kontrol akses minimum terdokumentasi.
- Ada bukti integrasi data/aksi sesuai konteks modul.
- Ada dokumentasi handover dan catatan known issues/next steps.

9. Non-Functional Requirements

NFR ID	Category	Requirement
NFR-001	Security	Sistem harus menerapkan autentikasi, role-based authorization, dan audit logging.
NFR-002	Consistency	Data pelanggan harus konsisten di seluruh modul.
NFR-003	Usability	Antarmuka harus mudah digunakan oleh role bisnis yang berbeda.
NFR-004	Scalability	Struktur modul dan API harus memungkinkan penambahan fitur baru.
NFR-005	Observability	Error, event integrasi, dan perubahan data harus dapat dimonitor.
NFR-006	Traceability	Requirement harus dapat ditelusuri ke modul, output, dan review.

10. Traceability Matrix

Requirement ID	Module	Expected Deliverables
BR-001	Authentication & User Management	RE Document, UI/UX Spec, API Contract, Implementation Report, Test Notes, Demo & Handover
BR-002	Customer Data Management	RE Document, UI/UX Spec, API Contract, Implementation Report, Test Notes, Demo & Handover
BR-003	Sales Automation	RE Document, UI/UX Spec, API Contract, Implementation Report, Test Notes, Demo & Handover
BR-004	Analytics Dashboard & Reporting	RE Document, UI/UX Spec, API Contract, Implementation Report, Test Notes, Demo & Handover
BR-005	Team Collaboration & Communication	RE Document, UI/UX Spec, API Contract, Implementation Report, Test Notes, Demo & Handover
BR-006	Main Dashboard & Notifications	RE Document, UI/UX Spec, API Contract, Implementation Report, Test Notes, Demo & Handover
BR-007	System Integration & API	RE Document, UI/UX Spec, API Contract, Implementation Report, Test Notes, Demo & Handover
BR-008	Marketing Automation	RE Document, UI/UX Spec, API Contract, Implementation Report, Test Notes, Demo & Handover
BR-009	Customer Service System	RE Document, UI/UX Spec, API Contract, Implementation Report, Test Notes, Demo & Handover
BR-010	Customer Segmentation & Prediction	RE Document, UI/UX Spec, API Contract, Implementation Report, Test Notes, Demo & Handover

11. Dependencies and Risks

- Ketergantungan antar modul cukup tinggi, terutama Customer Data Management sebagai source utama data.
- Ketidakjelasan permission matrix dapat memengaruhi banyak modul sekaligus.
- Integrasi real-time dan notifikasi dapat menimbulkan kompleksitas teknis tambahan.
- Marketing Automation dan Customer Segmentation berpotensi meluas jika tidak dibatasi oleh baseline bisnis.
- Integrasi eksternal sebaiknya dikelola sebagai fase lanjutan.

12. Management Review Checklist

Review Area	Question	Status	Reviewer Notes
Scope	Apakah seluruh modul CRM sudah tercakup?		
Business Fit	Apakah kebutuhan bisnis sudah sesuai dengan tujuan SmartCRM79?		
Role Coverage	Apakah kebutuhan tiap role sudah tercermin?		
Flow Diagrams	Apakah flow global dan flow per modul cukup jelas?		
Deliverables	Apakah output minimum per modul memadai?		
Approval	Apakah sign-off pihak terkait sudah lengkap?		

13. Appendix A – Module Flow Catalogue

Authentication & User Management

[Start] -> [Login/Register] -> [MFA] -> [Session/Token] -> [Role & Permission Check] -> [Access Module] -> [Audit Log]

Customer Data Management

[Open Customer Module] -> [Create/Search Customer] -> [View/Edit Profile] -> [Tag/Segment/Attach File] -> [Save] -> [Timeline Updated]

Sales Automation

[Create/Import Lead] -> [Qualify Lead] -> [Move Opportunity Stage] -> [Auto Task & Reminder] -> [Proposal/Negotiation] -> [Win/Loss] -> [Forecast Updated]

Analytics Dashboard & Reporting

[Collect CRM Data] -> [Aggregate KPI] -> [Apply Filters] -> [Render Dashboard/Charts] -> [Export/Schedule] -> [Insight]

Team Collaboration & Communication

[Open Shared Customer Profile] -> [Add Note/Comment/Mention] -> [Status Update] -> [Notification Broadcast] -> [Calendar Follow-up] -> [Audit Trail]

Main Dashboard & Notifications

[Login Success] -> [Load Role-based Dashboard] -> [Show Notifications] -> [Show Tasks/Deadlines] -> [Navigate via Quick Links]

System Integration & API

[Module Event Triggered] -> [API Service] -> [Validation/Security Check] -> [Sync Module] -> [Log/Monitor] -> [Optional Webhook]

Marketing Automation

[Create Campaign] -> [Design Template] -> [Set Segment/Trigger] -> [Schedule/Run] -> [Track Engagement] -> [Update Lead Score]

Customer Service System

[Create Ticket] -> [Auto Assign] -> [Work on Ticket] -> [SLA Monitor/Escalate] -> [Resolve] -> [Collect Feedback]

Customer Segmentation & Prediction

[Collect Activity/Transaction Data] -> [RFM Calculation] -> [Segment Assignment] -> [Predict Purchase/Churn] -> [Generate Insight] -> [Recommendation]

14. Appendix B – Output Types Baseline

Output Type	Definition
RE Document	Analisis referensi/As-Is, gap, dan kebutuhan To-Be.
UI/UX Spec	Wireframe, state UI, validasi, dan behavior interface.
API Contract	Endpoint, request/response, error code, dan aturan integrasi.
Implementation Report	Bukti implementasi FE/BE, PR/commit, dan catatan teknis.
Test Notes	Skenario happy path dan edge case beserta hasil aktual.
Demo & Handover	Demo fitur, known issues, dan dokumentasi handover.

End of Document